

Module 24.14: Logistic Regression — Cost Function & Optimization

1. Why Mean Squared Error (MSE) Fails for Logistic Regression

In linear regression, MSE yields a strictly convex paraboloid. However, if we insert the non-linear Sigmoid hypothesis $\hat{y} = \sigma(\mathbf{w}^T \mathbf{x} + b)$ into the MSE loss:

$$J_{\text{MSE}}(\mathbf{w}) = \frac{1}{2m} \sum_{i=1}^m \left(\sigma(\mathbf{w}^T \mathbf{x}^{(i)} + b) - y^{(i)} \right)^2$$

The resulting loss surface is **non-convex**, riddled with flat plateaus and local minima! Gradient descent gets trapped and fails to converge to the optimal parameters.

2. Derivation of the Binary Cross-Entropy Cost via Maximum Likelihood Estimation (MLE)

For binary targets $y \in \{0, 1\}$, the outcome follows a **Bernoulli distribution**:

$$P(y|\mathbf{x}) = \hat{y}^y (1 - \hat{y})^{1-y}$$

For a dataset of m independent observations, the joint **Likelihood Function** is:

$$L(\mathbf{w}, b) = \prod_{i=1}^m P(y^{(i)}|\mathbf{x}^{(i)}) = \prod_{i=1}^m (\hat{y}^{(i)})^{y^{(i)}} (1 - \hat{y}^{(i)})^{1-y^{(i)}}$$

Taking the Log-Likelihood:

$$\ln L(\mathbf{w}, b) = \sum_{i=1}^m \left[y^{(i)} \ln \hat{y}^{(i)} + (1 - y^{(i)}) \ln(1 - \hat{y}^{(i)}) \right]$$

Converting to Minimization (Negative Log-Likelihood / Log Loss):

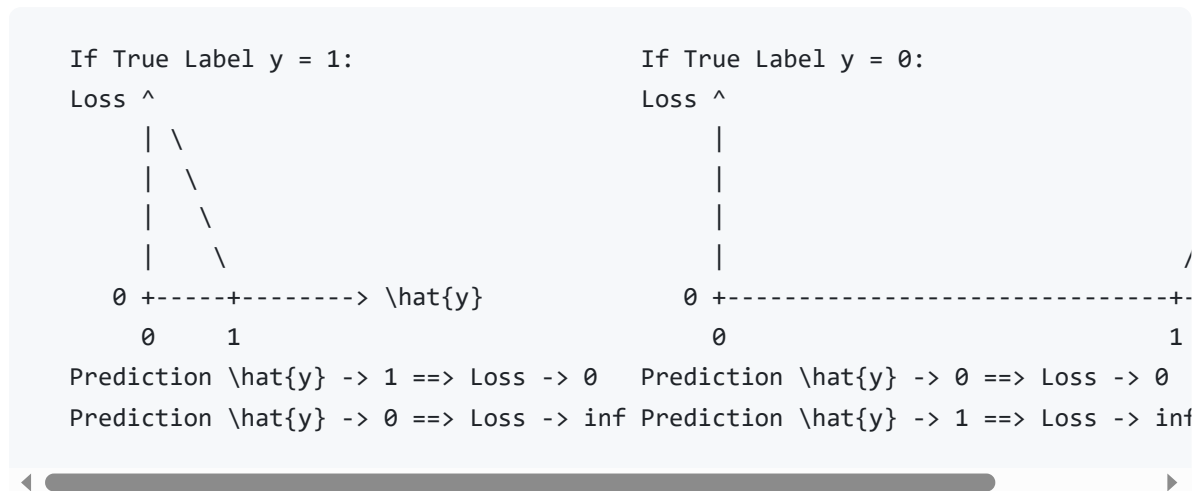
Dividing by $-m$, we obtain the **Binary Cross-Entropy (Log Loss)** cost function:

$$J(\mathbf{w}, b) = -\frac{1}{m} \sum_{i=1}^m \left[y^{(i)} \ln \hat{y}^{(i)} + (1 - y^{(i)}) \ln(1 - \hat{y}^{(i)}) \right]$$

3. Piecewise Penalty Mechanics

The loss per sample $L(\hat{y}, y)$ exhibits asymmetric logarithmic penalties:

$$L(\hat{y}, y) = \begin{cases} -\ln(\hat{y}) & \text{if } y = 1 \\ -\ln(1 - \hat{y}) & \text{if } y = 0 \end{cases}$$



If the model predicts confident wrong answers ($y = 1$ but $\hat{y} = 0$), the penalty approaches **infinity**!

4. Analytic Gradient Derivation & Update Rule

Applying the chain rule using $\frac{d\sigma}{dz} = \sigma(z)(1 - \sigma(z))$:

$$\frac{\partial J}{\partial w_j} = \frac{1}{m} \sum_{i=1}^m \left(\hat{y}^{(i)} - y^{(i)} \right) x_j^{(i)}$$

$$\frac{\partial J}{\partial b} = \frac{1}{m} \sum_{i=1}^m \left(\hat{y}^{(i)} - y^{(i)} \right)$$

Vectorized Gradient Descent Update:

$$\mathbf{w} := \mathbf{w} - \alpha \left[\frac{1}{m} \mathbf{X}^T (\hat{\mathbf{y}} - \mathbf{y}) \right]$$

$$b := b - \alpha \left[\frac{1}{m} \sum_{i=1}^m (\hat{y}^{(i)} - y^{(i)}) \right]$$